

2 What is Data? 什么是数据

“Data Mining is the study of collecting, processing, analyzing, and gaining useful insights from data” – Charu Aggarwal

数据挖掘的含义：“数据挖掘是一门关于数据收集、处理、分析并从中获取有用洞见的学科”——查鲁·阿加瓦尔

Data → Data Mining → Value

Data

- Collection of data objects and their attributes
数据对象及其属性的集合。
- An **attribute** is a property or characteristic of an object (**columns**)
属性是对象的一个性质或特征。
 - Examples: name, date of birth, height, occupation.
示例：姓名、出生日期、身高、职业。
 - Attribute is also known as variable, field, characteristic, or feature
属性也被称为变量、字段、特性或特征。
- For each object the attributes take some values.
每个对象的属性都拥有特定数值。
- The collection of attribute-value pairs describes a specific object
属性-值对的集合用于描述特定对象。
 - Object is also known as record, point, case, sample, entity, or instance
对象也被称为记录、点、案例、样本、实体或实例。
- **Size (n):** Number of objects (**rows**)
大小 (n) : 对象 (行) 的数量
- **Dimensionality (d):** Number of attributes
维度 (d) : 属性数量
- **Density/Sparsity:** Number of populated object-attribute pairs
密度/稀疏度：已填充对象-属性对的数量

Attributes				
Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Objects

Relational Data 关系型数据

- The term comes from **DataBases**, where we assume data is stored in a **relational table** with a fixed schema (fixed set of attributes)

该术语源自数据库领域，在此领域中我们假定数据存储于具有固定模式（即固定属性集）的**关系表**中。

- In Databases, it is usually assumed that the table is dense (few null values)

在数据库中，通常假设表是密集的（少有null值）。

- There are a lot of data in this form
 - E.g., census data 普查数据
- There are also a lot of data which do not fit well in this form

此外，有大量数据并不完全符合此格式。

- Sparse data: Many missing values
松散度高的数据
- Not easy to define a fixed schema
数据没有固定的模式

Types of Attributes 属性的类型

There are different types of attributes

存在多种属性类型。

- Numeric 数值型
 - Examples: dates, temperature, time, length, value, count.
实例：日期、温度、时间、长度、数值、计数。
 - Discrete** (counts) vs **Continuous** (temperature)
离散型（计数）与连续型（温度）
 - Special case: **Binary/Boolean** attributes (yes/no, exists/not exists)

特殊情况：二进制/布尔属性（是/否，存在/不存在）

- Categorical

分类的

- Examples: eye color, zip codes, strings, rankings (e.g, good, fair, bad), height in {tall, medium, short}

示例：眼睛颜色、邮政编码、字符串、等级（例如：优秀、良好、差）、身高等级（高、中等、矮）

- **Nominal** (no order or comparison) vs **Ordinal** (order but not comparable)

名义尺度（无顺序或不可比较）与顺序尺度（有顺序但不可比较）

Numeric Relational Data 数值型关系数据

If data objects have the same **fixed set** of **numeric attributes**, then the data objects can be thought of as **points/vectors** in a multi-dimensional space, where each **dimension** represents a distinct attribute

若数据对象拥有相同的**固定**数值属性集，则可将这些数据对象视作多维空间中的**点或向量**，其中每一**维度**代表一个不同的属性。

Such data set can be represented by an **n-by-d data matrix**, where there are **n** rows, one for each object, and **d** columns, one for each attribute

该数据集可由一个 $n \times d$ 的数据矩阵表示，其中包含 n 行（每行代表一个对象）和 d 列（每列对应一个属性）。

Numeric Data 数值数据

- Thinking of numeric data as **points or vectors** is very convenient

将数值数据视为点或向量极为便利。

- For small dimensions we can plot the data

对于小维度数据，我们可以进行绘图展示。

- We can use geometric analogues to define concepts like **distance** or **similarity**

我们可以运用几何类比来定义诸如距离或相似度这样的概念。

- We can use **linear algebra** to process the data matrix

我们可以利用线性代数来处理数据矩阵。

Vector Databases 向量数据

- This is a special case of numerical data, where each object is a multidimensional numerical vector.

这是数值数据中的一个特例，其中每个对象均为多维数值向量。或者说是一个字段里存放多个数值，组成一个向量

- Different from before, the vectors do not store the attribute values for the object, but rather the **embedding** of the object as it is produced by some Machine Learning algorithm.

与以往不同，这些向量并非存储对象的属性值，而是保存由某种机器学习算法生成的对象**嵌入**表示。

- Embedding vectors are **fully dense** and have no clear interpretation.

嵌入向量是完全密集的且不具备明确的可解释性。

- They have been shown to be remarkably successful in capture **semantic relationships** between objects.

它们已被证明在捕捉物体间的**语义关联**方面成效显著。

Categorical Relational Data 分类关系数据

Data that consists of a collection of records, each of which consists of a **fixed set** of **categorical** attributes

数据由一系列记录组成，其中每条记录包含一组固定的分类属性。

Mixed Relational Data 混合关系数据

Data that consists of a collection of records, each of which consists of a fixed set of both **numeric** and **categorical** attributes

数据由一组记录构成，其中每条记录均包含固定数量的**数值型**与**分类型**属性。

需要注意的是，有些时候虽然有些属性中存放的数据是数字，但是这不代表该属性存储的就是数值型数据。

Boolean attributes can be thought as both numeric and categorical When appearing together with other attributes they make more sense as categorical They are often represented as numeric though

布尔属性既可被视为数值型，也可被视为分类型。当与其他属性同时出现时，它们作为分类属性更具意义，然而在实际应用中通常以数值形式进行表示。

Sometimes it is convenient to represent categorical attributes as boolean.

将分类属性表示为布尔值有时更为便捷。

- Add a Boolean attribute for each possible value of the attribute

为属性的每个可能值添加一个布尔属性。

ID Number	Zip Code	Age	Marital Status	Income	Income Bracket	Refund
1129842	45221	55	Single	250000	High	No
2342345	45223	25	Married	30000	Low	Yes
1234542	45221	45	Divorced	200000	High	No
1243535	45224	43	Single	150000	Medium	No

VS.

ID	Zip 45221	Zip 45223	Zip 45224	Age	Single	Married	Divorced	Income	Refund
1129842	1	0	0	55	0	0	0	250000	0
2342345	0	1	0	25	0	1	0	30000	1
1234542	1	0	0	45	0	0	1	200000	0
1243535	0	0	1	43	0	0	0	150000	0

Sometimes it is convenient to represent numerical attributes as **categorical**.

有时将数值属性表示为分类变量更为便捷。

- Group the values of the numerical attributes into **bins**

将数值属性的值分组到各个区间中。

ID Number	Zip Code	Age	Marital Status	Income	Income Bracket	Refund
1129842	45221	50s	Single	High	High	0
2342345	45223	20s	Married	Low	Low	1
1234542	45221	40s	Divorced	High	High	0
1243535	45224	40s	Single	Medium	Medium	0

Binning 分区

- Idea: split the range of the domain of the numerical attribute into bins (intervals).

概念：将数值属性的定义域范围划分为若干区间（或称分箱）。

- Every bucket defines a categorical value

每个桶定义一个分类值。



- How do we decide the number of bins?

我们如何确定分组数量？

- Depends on the granularity of the data that we want

取决于我们所需数据的粒度

- Sometimes domain knowledge is also used

有时也会用到领域知识。

Bucketization 分桶化

- How do we decide the size of the bucket?

我们如何决定桶的大小？

- Depends on the data and our application

取决于数据本身和我们的实际应用

- Equi-width bins:** All bins have the same size

等宽分箱：所有箱体具有相同的宽度。

- Example: split time into decades

例如：以十年为基准分割时间

- Problem: some bins may be very sparse or empty

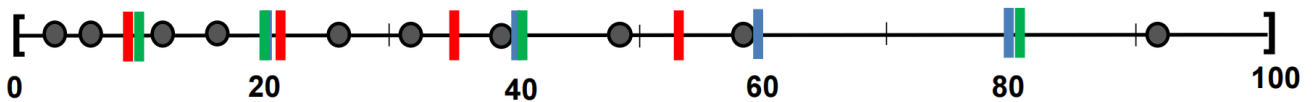
但是有些桶可能数据分布非常松散或者为空

- Equi-size (depth) bins:** Select the bins so that they all contain the same number of elements

等深分箱法：选择分箱方式，使得每个箱中包含相同数量的元素。

- This splits data into quantiles: top-10%, second 10% etc
将数据按分位数划分：前10%、第二个10%等。
 - Some bins may be very small
某些区域可能非常小
- **Equi-log bins:** $\log \text{end} - \log \text{start}$ is constant
等对数区间：对数结束值减去对数起始值保持恒定。
 - The size of the previous bin is a fraction of the current one
前一个箱体的大小是当前箱体的一部分。
 - Better for skewed distributions
更适合偏态分布
- **Optimized bins:** Use a 1-dimensional clustering algorithm to create the bins
优化分组：采用一维聚类算法创建分组

Example



Blue: Equi-width [20,40,60,80]

Red: Equi-depth (2 points per bin)

Green: Equi-log ($\frac{\text{end}}{\text{start}} = 2$)

Physical Data Storage 物理数据存储

- Stored in a **Relational Database**
存储于**关系型数据库**中
 - Assumes a strict **schema** and relatively **dense** data (few missing/Null values)
假设采用严格的模式且数据相对密集（缺失/空值较少）。
- **Tab or Comma separated files** (TSV/CSV), Excel sheets, relational tables
制表符或逗号分隔文件（TSV/CSV）、Excel表格、关系表
 - Assumes a strict schema and relatively dense data (few missing/Null values)
假设采用严格的数据模式且数据相对密集（缺失值或空值较少）
- Flat file with triplets (record id, attribute, attribute value)
包含三元组（记录标识符、属性、属性值）的平面文件
 - A very flexible data format, allows multiple values for the same attribute (e.g., phone number)
一种非常灵活的数据格式，允许同一属性（例如电话号码）拥有多个值。
- JSON, XML format
 - Standards for data description that are more flexible than relational tables
比关系表更灵活的数据描述标准

- There exist parsers for reading such data.

存在用于读取此类数据的解析器。

Comma Separated File

```
id,Name,Surname,Age,Zip
1,John,Smith,25,10021
2,Mary,Jones,50,96107
3,Joe,Doe,80,80235
```

- Can be processed with simple parsers, or loaded to excel or a database

Triple-store

```
1, Name, John
1, Surname, Smith
1, Age, 25
1, Zip, 10021
2, Name, Mary
2, Surname, Jones
2, Age, 50
2, Zip, 96107
3, Name, Joe
3, Surname, Doe
3, Age, 80
3, Zip, 80235
```

- Easy to deal with missing values

JSON EXAMPLE – Record of a person

```
{
  "firstName": "John",
  "lastName": "Smith",
  "isAlive": true,
  "age": 25,
  "address": {
    "streetAddress": "21 2nd Street",
    "city": "New York",
    "state": "NY",
    "postalCode": "10021-3100"
  },
  "phoneNumbers": [
    {
      "type": "home",
      "number": "212 555-1234"
    },
    {
      "type": "office",
      "number": "646 555-4567"
    }
  ],
  "children": [],
  "spouse": null
}
```

XML EXAMPLE – Record of a person

```
<person>
  <firstName>John</firstName>
  <lastName>Smith</lastName>
  <age>25</age>
  <address>
    <streetAddress>21 2nd
Street</streetAddress>
    <city>New York</city>
    <state>NY</state>
    <postalCode>10021</postalCode>
  </address>
  <phoneNumbers>
    <phoneNumber>
      <type>home</type>
      <number>212 555-1234</number>
    </phoneNumber>
    <phoneNumber>
      <type>fax</type>
      <number>646 555-4567</number>
    </phoneNumber>
  </phoneNumbers>
  <gender>
    <type>male</type>
  </gender>
</person>
```

JSON Databases JSON数据格式

- There are specialized Databases for semi-structured data such as

存在专为半结构化数据设计的数据库类型，例如

JSON data

- Commonly used DBs:

- MongoDB
 - CouchDB
 - MySQL
- The query language is very similar to SQL used for relational data.

查询语言与用于关系数据的SQL极为相似。

Beyond Relational Data: Set Data 超越关系型数据：集合数据

- Each record is a set of items from a space of possible items

每条记录都是来自可能项空间中的一个项集。

- Example: Transaction data

例如：交易数据

- Also called market-basket data
亦称为市场篮子数据。

- Example: Document data

示例：文档数据

- Also called bag-of-words representation
亦称为词袋表示法。

Vector representation of market-basket data 市场篮子数据的矢量表示

- Market-basket data can be represented, or thought of, as numeric vector data

市场篮子数据可以被表示或视为数值向量数据。

- The vector is defined over the set of all possible items
该向量定义在所有可能的项目集合上。
- The values are binary (the item appears or not in the set)
这些值是二元的（项目在集合中出现或不出现）。

TID	Items
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

TID	Bread	Coke	Milk	Beer	Diaper
1	1	1	1	0	0
2	1	0	0	1	0
3	0	1	1	1	1
4	1	0	1	1	1
5	0	1	1	0	1

Sparsity: Most entries are zero. Most baskets contain few items

稀疏性：大多数条目为零。大多数购物篮中仅含少量商品。

Vector representation of document data 文档数据的向量表示

- Document data can be represented, or thought of, as numeric vector data
文档数据可以表示为，或被视为数值向量数据。
 - The vector is defined over the set of all possible words
向量是在所有可能词汇的集合上定义的。
 - The values are the counts (number of times a word appears in the document)
数值代表词频（即某个词汇在文档中出现的次数）。

Doc Id	Words
1	the, dog, follows, the, cat
2	the, cat, chases, the, cat
3	the, man, walks, the, dog

Doc Id	the	dog	follows	cat	chases	man	walks
1	2	1	1	1	0	0	0
2	2	0	0	2	1	0	0
3	1	1	0	0	0	1	1

Sparsity: Most entries are zero. Most documents contain few of the words

稀疏性：大多数条目为零。大多数文档仅包含少量词汇。

Physical Data Storage 物理数据存储

- Usually set data is stored in flat files
通常，集合数据存储在平面文件中。
 - One line per set
每行一组

Dependent Data 依赖数据

- In tables we usually consider each object independent of each other.
在表格中，我们通常认为各个对象彼此独立。
- In some cases, there are explicit **dependencies** between the data
在某些情况下，数据之间存在着明确的依赖关系。
 - Ordered/Temporal data**: We know the time order of the data
有序/时序数据：我们知道数据的时间顺序。
 - Spatial data**: Data that is placed on specific locations
空间数据：位于特定地理位置的数据。
 - Spatiotemporal data**: data with location and time
时空数据：包含位置和时间信息的数据
 - Networked/Graph data**: data with pairwise relationships between entities
网络/图数据：指实体间存在成对关系的数据。

Ordered Data 有序数据

- Genomic sequence data 基因组序列数据
- Data is a long ordered string 数据是一个长的有序字符串
- Time series 时间序列
 - Sequence of ordered (over “time”) numeric values.
按时间顺序排列的数值序列。
- Sequence data: Similar to the time series but in this case, we have categorical values rather than numerical ones.

序列数据：与时间序列类似，但此处涉及的是分类值而非数值型数据。

- Example: Event logs
例如：事件日志

Spatial Data 空间数据

- Attribute values that can be arranged with geographic co-ordinates
可依据地理坐标进行排列的属性值
 - Measurements of temperature/pressure in different locations.
不同位置的温度/压力测量。
 - Sales numbers in different stores
各门店销售数据
 - The majority party in the country states (categorical)
该国多数政党（明确地）声明
- Such data can be nicely visualized.
此类数据可以很好地可视化。

Spatiotemporal Data 时空数据

- Data that have both spatial and temporal aspects
具有时空双重属性的数据
 - Measurements in **different locations over time**
随时间推移在不同地点进行的测量
 - Pressure, Temperature, Humidity
压力，温度，湿度
 - Measurements that **move in space over time**
随空间位置变化而动态测量的数据
 - Traffic, **Trajectories** of moving objects
交通：移动物体的轨迹

Graph Data 图数据

- Graph data: a collection of **entities** and their **pairwise relationships**.

图数据：一组实体及其两两关系的集合。

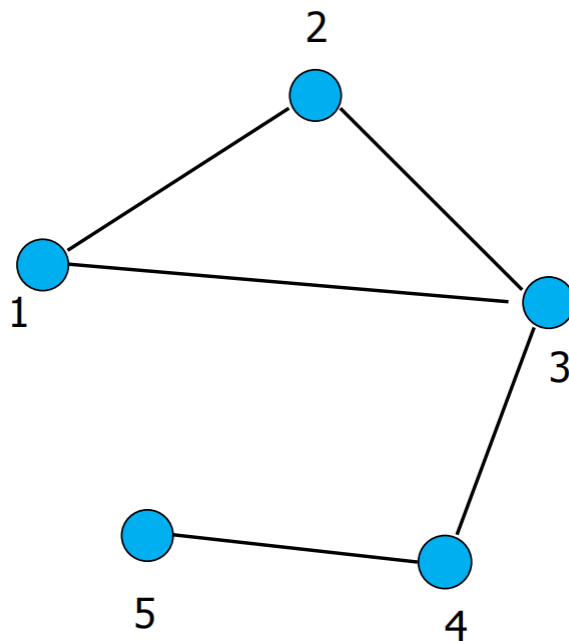
- Examples:

- Web pages and hyperlinks
网页与超链接
- Facebook users and friendships
Facebook用户与好友关系
- The connections between brain neurons
大脑神经元之间的连接
- Genes that regulate each other
调控彼此的基因

- In this case the data consists of **pairs**:

在这种情况下，数据由成对组成：

- Who links to whom
谁链接到谁
- We may have **undirected** links
我们可能拥有无向链接。

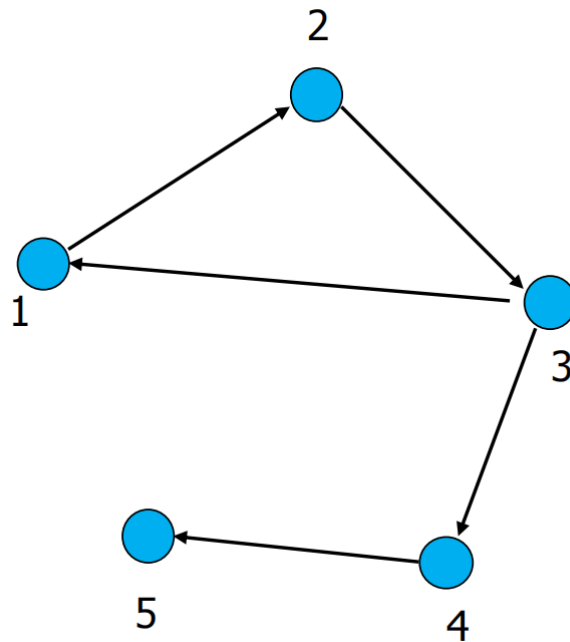


- In this case the data consists of **pairs**:

在这种情况下，数据由成对组成：

- Who links to whom
谁链接到谁

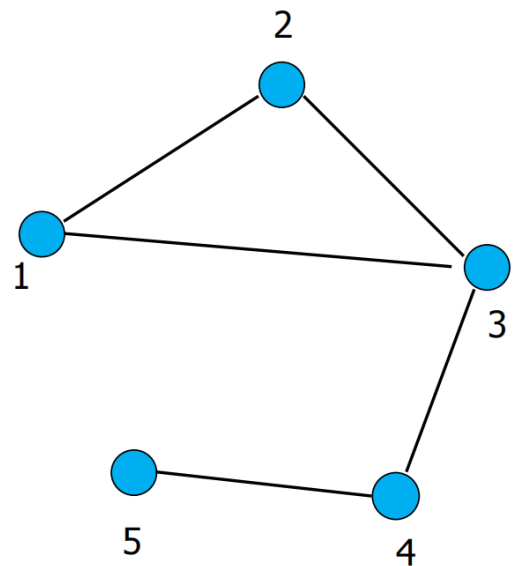
- Or **directed** links
可能是有向的连接



Representation

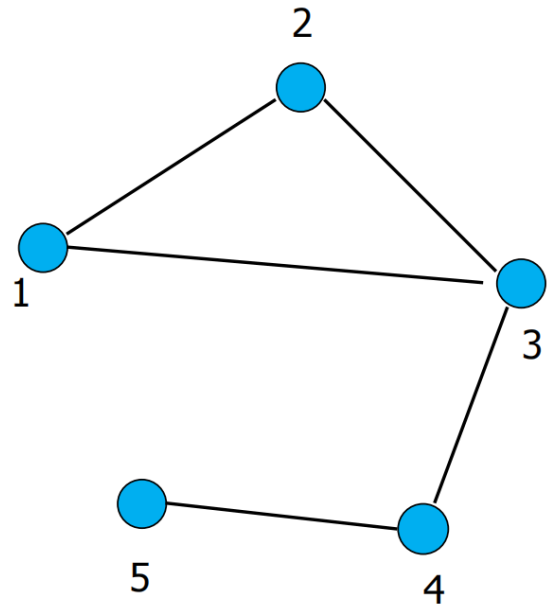
- Adjacency matrix 邻接矩阵
 - Very sparse, very wasteful, but useful conceptually
极为稀疏，极其浪费，但在概念层面上却颇为实用。

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$



- Adjacency list
邻接列表
 - Not so easy to maintain
不容易维护

1: [2, 3]
2: [1, 3]
3: [1, 2, 4]
4: [3, 5]
5: [4]



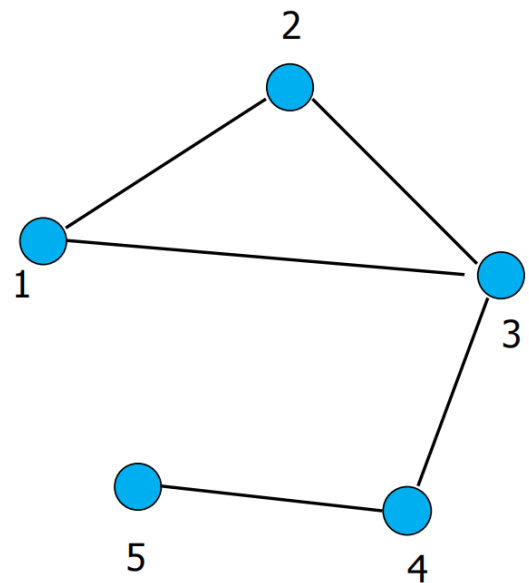
- List of pairs

配对列表

- The simplest and most efficient representation

最简单且高效的表达方式

(1,2)
(2,3)
(1,3)
(3,4)
(4,5)



Types of data: summary

- Numeric data: Each object is a point in a multidimensional space

数值数据：每个对象均为多维空间中的一个点。

- Categorical data: Each object is a vector of categorical values

分类数据：每个对象都是一个分类值向量。

- Set data: Each object is a set of values (with or without counts)

数据集：每个对象为一组值（可含计数或不含）。

- Sets can also be represented as binary vectors, or vectors of counts

集合也可以用二进制向量或计数向量来表示。

- Dependent data:

相关数据:

- Ordered sequences: Each object is an ordered sequence of values.

有序序列：每个对象都是一个值的有序序列。

- Spatial data: objects are fixed on specific geographic locations

空间数据：对象固定于特定的地理位置。

- Graph data: A collection of pairwise relationships

图数据：成对关系的集合

- The data matrix:

数据矩阵

- In almost all types of data we can find a way to transform the data into a matrix, where the rows correspond to different records, and the columns to numeric attributes

几乎在所有类型的数据中，我们都能找到将数据转换为矩阵的方法，其中行对应不同的记录，列对应数值属性。